The Ghoshian Six-Body Problem

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Abstract

This treatise develops a proposed interdisciplinary framework for coordinating six interdependent institutional functions under planetary-scale uncertainty. Its motivating sequence is local convergence among neighbouring central banks, planetary coordination among heterogeneous regional systems, and the possibility of still higher-order coordination across a Solar System. The central thought experiment is an external existential threat to Earth, represented without assuming that the origin of a signal establishes the intentions of every entity associated with that origin. The framework combines institutional network theory, Bayesian persuasion and information design, mechanism design, welfare economics, central banking, public finance, law, risk management, astronomy, physics, biology, psychology, political science, diplomacy, computer science, statistics, econometrics, cybersecurity and AI.

The phrase "six-body problem" is used here as an institutional and computational analogy, not as a claim that a classical six-body gravitational system has been solved.

The arithmetic identity $6 = 1+2+3$ and the triangular-number identity $6 = 3(3+1)/2$ motivate a six-node architecture but do not prove its optimality. The six nodes proposed here are: (i) observation and scientific assessment; (ii) verification, intelligence and epistemic security; (iii) deliberation, diplomacy and lawful authorisation; (iv) monetary, financial and resource coordination; (v) logistics, civil protection and operational response; and (vi) welfare, rights, audit and intergenerational oversight. These are functional roles, not necessarily six sovereign institutions or six independent legal persons.

The principal thesis is that planetary coordination need not require institutional uniformity. A legitimate information architecture can help heterogeneous agents update beliefs and choose compatible actions while preserving pluralism, sovereignty, accountability and individual rights. Planetary Bayesian Persuasion (PBP) is introduced as a research programme: a constrained, multi-receiver extension of Bayesian information design in which the objective is not merely to persuade, but to support robust, welfare-sensitive and publicly accountable coordination. Novel constructions are identified as proposals or hypotheses requiring formal and empirical validation.

The treatise ends with "The End"

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1 Introduction

1.1 The three scales of convergence

The motivating programme distinguishes three nested scales:

1. **Local convergence:** coordination or integration among neighbouring central banks and monetary institutions.
2. **Planetary convergence:** coordination among regional monetary, political, scientific and operational systems.
3. **Solar System convergence:** coordination among planetary-scale systems and any future extraterrestrial settlements or institutions.

These are analytically distinct. Local coordination does not logically entail planetary coordination, and planetary coordination does not entail a single world government, currency or central bank.

The user's treatise on the convergence of central banks of neighbouring nations identifies interest-rate synchronisation, exchange-rate management, financial-stability measures and regulatory harmonisation as dimensions of local convergence. It also identifies trade, financial and exchange-rate channels as transmission mechanisms, and recognises political, legal and operational barriers to deeper integration [1]. This work takes that local framework as a starting point, not as empirical proof that planetary convergence is inevitable.

1.2 The welfare question

The question "welfare of whom?" cannot be answered by naming only states, central banks or governments. A planetary framework must identify all affected populations: individuals, national communities, regional populations, humanity as a whole, future generations and, where relevant, non-human sentient stakeholders. Aggregate welfare can conceal severe losses imposed on minorities or distant populations. Distribution, rights and uncertainty must therefore remain explicit.

The governing principle is:

$$\boxed{\text{Institutional convergence is a possible mechanism; welfare is the criterion.}} \tag{1}$$

This principle does not prescribe a fixed institutional count. Nor does it imply that all functions should be merged. Shared information, interoperable systems and joint contingency planning may coexist with legally independent institutions.

1.3 Epistemic status

Four categories are distinguished:

Established results: standard mathematical, economic or scientific results supported by cited literature.

Source-derived propositions: concepts and claims stated in preceding user-authored treatises, represented here in their original scope.

Model constructions: definitions and mathematical structures introduced here.

Empirical hypotheses: claims requiring data, identification and falsification.

The Jupiter-originating alien-force scenario is a thought experiment, not an assertion about an actual threat.

2 The arithmetic and architecture of six

2.1 The numerical motif

The numerical motivation is:

$$6 = 1 + 2 + 3, \tag{2}$$

$$6 = \frac{3(3+1)}{2} = T_3. \tag{3}$$

Equation (3) identifies six as the third triangular number. Neither identity establishes that six is a uniquely optimal number of institutional functions. Six is adopted as a design hypothesis to be compared with alternatives.

2.2 Six functional bodies

Let the six functional bodies be

$$V = \{O, V, D, F, L, W\}. \tag{4}$$

Their proposed roles are:

O observation, astronomy, sensing and scientific assessment;

V verification, intelligence, provenance and epistemic security;

D deliberation, diplomacy, political authorisation and lawful decision;

F monetary, financial, fiscal and resource coordination;

L logistics, civil protection, infrastructure and operational response;

W welfare, rights, audit, accountability and intergenerational oversight.

These are functional roles. One organisation may perform more than one role, and one role may be distributed across many organisations. This distinction prevents a category error between six functions and six legal institutions.

2.3 Institutional graph

Represent the architecture by a directed, weighted, time-varying graph:

$$G_t = (V, E_t, A_t), \tag{5}$$

where $A_t = [a_{ij,t}]$ records the intensity and direction of information, authority, resource or operational dependencies. The graph is multiplex: an edge may carry information without carrying command authority, or resources without transferring legal competence.

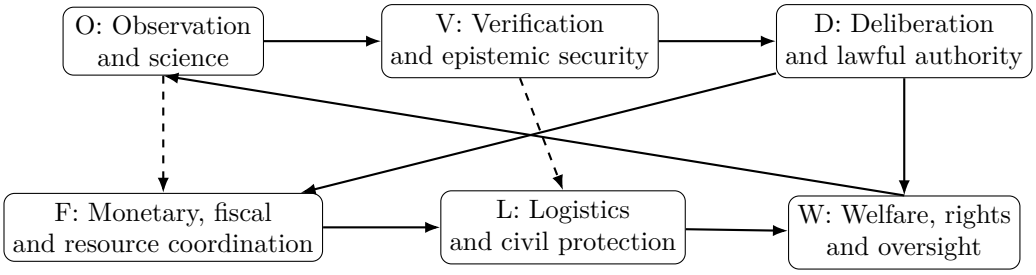


Figure 1: A proposed six-function planetary coordination architecture. Arrows illustrate possible information and feedback pathways, not a prescribed chain of command.

2.4 The coupled decision system

Let $x_{i,t}$ denote the state of body i, $a_{i,t}$ its action and $\varepsilon_{i,t}$ a disturbance. A general coupled system is

$$x_{i,t+1} = f_i\left(x_{i,t}, a_{i,t}, \{x_{j,t}, a_{j,t}\}_{j\neq i}, \varepsilon_{i,t}\right). \tag{6}$$

The system is not solved by solving six isolated optimisation problems. Interdependence means each action changes the feasible set, information or payoff of other bodies. The analytical task is to establish existence, stability, robustness and welfare properties under explicit assumptions.

3 From local monetary convergence to planetary coordination

3.1 Local convergence metrics

For n neighbouring central banks with policy rates $r_{i,t}$, a descriptive cross-sectional dispersion measure is

$$\sigma_{r,t} = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(r_{i,t} - \bar{r}_t)^2}, \qquad \bar{r}_t = \frac{1}{n}\sum_{i=1}^{n} r_{i,t}. \tag{7}$$

A falling $\sigma_{r,t}$ is evidence of rate convergence in this narrow descriptive sense. It is not, by itself, evidence of welfare improvement, causal policy coordination or institutional convergence. Common shocks can synchronise rates without deliberate coordination.

Institutional convergence may be represented by a vector of similarities in inflation targeting, communication, financial-stability mandates and governance:

$$ICI_t = \omega_1 IT_t + \omega_2 CS_t + \omega_3 FS_t + \omega_4 GF_t, \qquad \omega_k \geq 0, \quad \sum_{k=1}^{4}\omega_k = 1. \tag{8}$$

This index is meaningful only after each component is operationally defined and measurement invariance across jurisdictions is assessed.

3.2 Regional systems as planetary agents

Let

$$\mathcal{R} = \{R_1, \ldots, R_n\} \tag{9}$$

denote regional systems. Each R_i contains institutions, populations, legal competences, financial networks and policy preferences. Planetary coordination is not the forced identity $R_i = R_j$. It is the design of interfaces and agreements that make selected actions mutually compatible.

A regional state may be represented as

$$z_{i,t} = (m_{i,t}, g_{i,t}, c_{i,t}, k_{i,t}, q_{i,t}), \tag{10}$$

where components may encode monetary conditions, governance, cognitive and scientific capacity, critical infrastructure and risk exposure.

3.3 Compatibility rather than uniformity

Define

$$\mathcal{C}_{ij}(a_i, a_j, z_i, z_j) \in [0, 1], \tag{11}$$

where higher values indicate greater interoperability or consistency between regional actions. Compatibility is not identical policy. Two central banks may retain different policy rates while agreeing on swap-line protocols, settlement standards, crisis communications and supervisory information exchange.

4 Bayesian foundations

4.1 States, priors and signals

Let Θ be a finite or measurable set of states of the world. A receiver has prior belief $\mu_i(\theta)$. A sender commits to an information structure $\pi(s \mid \theta)$, generating signal s conditional on state θ. Upon observing s, receiver i updates by Bayes' rule:

$$\mu_i(\theta \mid s) = \frac{\pi_i(s \mid \theta)\mu_i(\theta)}{\sum_{\theta' \in \Theta} \pi_i(s \mid \theta')\mu_i(\theta')}. \tag{12}$$

The receiver chooses an action $a_i \in A_i$ to maximise expected utility:

$$a_i^*(s) \in \arg\max_{a_i \in A_i} \sum_{\theta \in \Theta} \mu_i(\theta \mid s) u_i(a_i, \theta). \tag{13}$$

4.2 Bayes plausibility

For a common prior and correctly specified signal structure, posterior beliefs average back to the prior:

$$\sum_s \Pr(s)\mu(\theta \mid s) = \mu(\theta). \tag{14}$$

Proposition 4.1 (Bayes plausibility)**.** *Suppose the sender commits to a signal rule $\pi(s \mid \theta)$ and the receiver updates by Bayes' rule under prior μ. Then the expectation of the posterior equals the prior, as in (14).*

Proof. By the law of total probability,

$$\sum_s \Pr(s)\mu(\theta \mid s) = \sum_s \Pr(s, \theta) \tag{15}$$

$$= \Pr(\theta) \tag{16}$$

$$= \mu(\theta). \tag{17}$$

Thus information can redistribute beliefs across realised signals, but cannot change the ex ante average belief when the prior and signal model are correct. □

This condition does not guarantee that the sender is truthful, that the model is correctly specified, or that receivers share the same prior. Those are separate institutional and epistemic questions.

4.3 Classical Bayesian persuasion

In the canonical model, a sender chooses an information structure to influence a receiver's action. The sender's objective may differ from the receiver's objective [2]. The present proposal generalises the setting to multiple receivers and adds explicit welfare, legitimacy and robustness constraints.

5 Planetary Bayesian Persuasion

5.1 Definition

Definition 5.1 (Planetary Bayesian Persuasion)**.** *Planetary Bayesian Persuasion (PBP) is a proposed multi-receiver information-design framework in which one or more senders choose or commit to information policies intended to support coordinated decisions among heterogeneous sovereign, regional and functional agents, subject to Bayes plausibility, epistemic integrity, lawful authority, rights protection, distributional welfare and robustness constraints.*

The term does not imply that planetary institutions should manipulate populations. Legitimate PBP concerns transparent organisation of evidence, uncertainty, provenance and decision-relevant communication. Persuasion is not coercion; neither is it a licence to deceive.

5.2 Multiple receivers

Let $i = 1, \ldots, n$ index regional or institutional receivers. Each may have a different prior μ_i, action set A_i, and utility u_i. A common public signal may be supplemented by institution-specific technical information:

$$a_i^*(s_i) \in \arg\max_{a_i \in A_i} \mathbb{E}_{\mu_i(\cdot|s_i)}[u_i(a_i, \theta, a_{-i})]. \tag{18}$$

Dependence on a_{-i} makes this a strategic information-design problem. A receiver's best response depends on expectations about other receivers' actions.

5.3 Planetary objective

A candidate objective is

$$\max_{\pi \in \mathcal{P}} \mathbb{E}_{\theta,s}\left[\sum_{i=1}^{n} w_i U_i(a_i^*(s_i), \theta) - \lambda_R \mathcal{R}(s, \theta) - \lambda_D \mathcal{D}(a)\right], \tag{19}$$

where w_i are explicit social weights, $\mathcal{R}$ measures epistemic or operational risk, $\mathcal{D}$ measures harmful incompatibility or coordination failure, and $\lambda_R, \lambda_D \geq 0$. The weights are normative choices, not scientific constants. A weighted sum must not conceal rights violations or catastrophic subgroup losses.

A more cautious formulation is constrained optimisation:

$$\begin{aligned} \max_{\pi \in \mathcal{P}} \quad & \mathbb{E}\left[\sum_i w_i U_i(a_i^*, \theta)\right], \\ \text{subject to} \quad & \Pr(\text{rights violation}) \leq \epsilon_R, \\ & \Pr(\text{catastrophic system failure}) \leq \epsilon_C, \\ & \mathbb{E}[\text{burden}_g] \leq B_g \quad \text{for protected group } g, \\ & \text{lawful authority and due process constraints hold,} \\ & \text{information provenance and audit requirements hold.} \end{aligned} \tag{20}$$

The thresholds require public justification and legal interpretation.

5.4 Pluralism and disagreement

A planetary mechanism should not presume that all receivers must reach the same posterior or select the same action. Different regions may rationally choose different responses because exposures, legal obligations and capabilities differ. The target is a feasible profile $a = (a_1, \ldots, a_n)$ satisfying shared constraints and acceptable compatibility, not uniformity.

Proposition 5.2 (Compatibility without uniformity)**.** *Suppose two receivers have distinct action sets and utilities. If there exist actions $a_1 \in A_1$ and $a_2 \in A_2$ satisfying shared safety and interoperability constraints, coordinated action is feasible without requiring $a_1 = a_2$.*

Proof. The constraints restrict the joint action profile, not equality of its components. A feasible profile may therefore contain different actions. This is a logical possibility result, not an empirical guarantee that such a profile exists in every application. □

6 The Jupiter-originating threat: a planetary stress test

6.1 Scenario definition

Consider a hypothetical signal indicating that an extraterrestrial force associated with Jupiter may threaten Earth. The scenario exposes epistemic, institutional and welfare failures. The location from

which a signal originates does not establish the intentions of Jupiter, all possible extraterrestrial actors, or any other celestial entity.

Let the uncertain state be

$$\Theta = \{\theta_H, \theta_N, \theta_U\}, \tag{21}$$

where θ_H denotes a hostile threat, θ_N a non-hostile explanation, and θ_U an unresolved state. The unresolved state prevents premature binary classification.

6.2 Observation and verification

Let y_t be observations from telescopes, spacecraft, remote sensors and other scientific instruments. A scientific observation model is

$$y_t = h(\theta, x_t) + \eta_t, \qquad \eta_t \sim p_\eta, \tag{22}$$

where x_t is the physical state and η_t represents measurement error. Independent replication, instrument calibration, provenance, time synchronisation and adversarial integrity checks are essential.

Verification evaluates whether observations are authentic, correctly interpreted and independently reproducible. It distinguishes measurement uncertainty, model uncertainty, source authenticity, strategic deception and genuine scientific disagreement.

6.3 Decision under uncertainty

Let a_i include monitoring, public warning, diplomatic contact, defensive preparation, infrastructure protection, evacuation planning and other lawful contingency measures. Expected loss is

$$\mathcal{L}_i(a_i) = \sum_{\theta \in \Theta} \mu_i(\theta \mid s) L_i(a_i, \theta). \tag{23}$$

The optimal action minimises expected loss subject to legal and ethical constraints. When probabilities are unreliable, robust or distributionally robust criteria may be preferable to a single posterior.

6.4 Avoiding the origin fallacy

$$\text{signal originates near Jupiter} \not\Rightarrow \text{Jupiter or all Jovian-associated entities are hostile.} \tag{24}$$

The source, agent, intention and capability must be separately established. The same distinction applies to human institutions: an individual actor must not be treated as representative of an entire population without evidence.

7 Welfare economics and the identity of the beneficiaries

7.1 Affected populations

Let $\mathcal{P}$ denote affected persons, partitioned into groups $g \in \mathcal{G}$. Let $w_p(a, \theta)$ be the welfare outcome for person p. A utilitarian aggregate is

$$W_U(a, \theta) = \sum_{p \in \mathcal{P}} \omega_p w_p(a, \theta), \qquad \omega_p \geq 0. \tag{25}$$

This is one normative approach, not a neutral fact. Alternatives include prioritarian, sufficientarian, capability-based, rights-constrained and multi-criteria evaluations [12, 11, 10].

7.2 Distributional analysis

Report at least:

$$\overline{\Delta W} = \frac{1}{|\mathcal{P}|} \sum_{p \in \mathcal{P}} \Delta w_p, \tag{26}$$

$$Q_\alpha(\Delta W) = \text{the } \alpha\text{-quantile of individual welfare changes}, \tag{27}$$

$$H_g = \Pr(\Delta w_p < -\tau_g \mid p \in g), \tag{28}$$

where τ_g is a substantively justified harm threshold. An increase in average welfare can coexist with severe losses for a subgroup. Reporting only the mean is inadequate.

7.3 Inside and outside institutional boundaries

A planetary institution affects members and non-members, present and future populations. Evaluation must include cross-border externalities, displaced risks, opportunity costs and compensation. Potential compensation is not actual compensation. The framework should report who pays, who benefits, who decides and who can appeal.

7.4 Intergenerational welfare

Let W_t be welfare at time t. A discounted objective is

$$W^{\text{intergen}} = \sum_{t=0}^{T} \delta^t W_t, \qquad 0 < \delta \leq 1. \tag{29}$$

The discount factor is ethically and economically consequential. Catastrophic irreversibility, ecological constraints and future generations' rights may require constraints rather than simple discounting.

8 Central banking, public finance and financial planning

8.1 Monetary institutions in a planetary emergency

The local convergence treatise identifies policy rates, exchange-rate arrangements, financial stability and regulatory harmonisation as important dimensions [1]. Under a planetary emergency, monetary institutions may additionally face payment continuity, liquidity provision, settlement resilience, capital-flow disruption, emergency financing and preservation of essential records.

The objective is not to subordinate central banks automatically to a planetary command. Pre-agreed protocols can clarify:

- which authority may activate emergency liquidity facilities;
- how collateral, solvency and fiscal backing are assessed;
- how cross-border swap lines and payment rails operate;
- how emergency measures expire and are audited;
- how monetary financing and fiscal transfers remain accountable.

8.2 Liquidity, solvency and real resources

Liquidity is the ability to meet obligations when due; solvency concerns the intertemporal capacity to meet liabilities. Emergency lending can address liquidity shortfalls but cannot make an insolvent entity solvent without loss allocation, recapitalisation or restructuring.

For institution i, a simplified balance-sheet identity is

$$A_{i,t} = L_{i,t} + E_{i,t}, \tag{30}$$

where A is assets, L liabilities and E equity. A planetary resource plan must distinguish monetary claims from real resources such as energy, food, transport, medicine, computing capacity and industrial inputs.

Let R_t be available real resources, C_t essential civilian consumption, D_t protective expenditure, I_t resilience investment and X_t other uses. Then:

$$C_t + D_t + I_t + X_t \leq R_t. \tag{31}$$

Money can mobilise resources but cannot by itself create scarce physical capacity. Financial planning should integrate cash-flow projections, contingent liabilities, scenario analysis, supply-chain bottlenecks and distributional incidence.

8.3 Avoiding financial domination

Emergency financial coordination must be temporary where appropriate, legally bounded, transparent and reviewable. Otherwise emergency arrangements can create permanent concentration of monetary and political power. Resilience requires redundancy, multiple payment pathways, auditable allocation rules and credible exit conditions.

9 Law, legitimacy and political authority

9.1 Legality, legitimacy and enforcement

A decision may be enforceable but unlawful, lawful but politically contested, or popular but inconsistent with rights. These dimensions must not be collapsed. An admissibility test separately examines legal authority and jurisdiction, legitimate authorisation, necessity and proportionality, due process and remedies, and time limits, transparency and independent audit.

9.2 Subsidiarity and multilevel governance

Subsidiarity suggests that decisions should be taken at the lowest level capable of addressing the problem effectively, while cross-border externalities may justify higher-level coordination. A planetary institution should exercise only competences that cannot be adequately discharged by local, national or regional bodies.

9.3 Emergency powers

Emergency powers require clear triggers, scope, duration, oversight and termination. A threat does not automatically suspend all legal protections. International law, human-rights law, humanitarian law and domestic constitutional law remain relevant; their precise application depends on facts and jurisdiction [25, 26, 27].

9.4 Persuasion and democratic consent

A public information system should disclose evidence, uncertainty, conflicts of interest, model assumptions and dissenting scientific views. It should not covertly manipulate beliefs or suppress lawful disagreement. Public persuasion is legitimate only when compatible with truthful communication, meaningful contestability and accountability.

10 Astronomy, physics and the limits of the gravitational analogy

10.1 Physical bodies and effective units

The preceding five-body astronomy treatise distinguishes five physical bodies from four effective external units after representing the Earth–Moon pair by its barycentre. The exact barycentric coordinate transformation reorganises the description; it does not destroy a physical degree of freedom. This is an analogy for institutional aggregation: a federation may be represented as one external actor while retaining internal dynamics.

For Earth and Moon positions $\mathbf{r}_E, \mathbf{r}_M$ and masses m_E, m_M, the barycentre and relative coordinate are

$$\mathbf{R}_{EM} = \frac{m_E \mathbf{r}_E + m_M \mathbf{r}_M}{m_E + m_M}, \tag{32}$$

$$\boldsymbol{\rho}_{EM} = \mathbf{r}_M - \mathbf{r}_E. \tag{33}$$

This is an exact coordinate change. Treating the pair as a single point mass for external dynamics may introduce approximation error depending on the external field and required precision.

10.2 Institutional gravitational pull

A preceding Ghoshian institutional proposal represents convergence pressure by

$$F_{\text{conv}} = k \frac{S_A - S_B}{D_{\text{inst}}^2}. \tag{34}$$

This is an analogy or candidate reduced-form relationship, not a physical law. Empirical use requires operational definitions of stability S, institutional distance D_{inst}, coefficient units, boundary behaviour and

possible asymmetry. The inverse-square denominator is not justified merely by analogy with Newtonian gravity.

A regularised candidate is

$$F_{\text{conv}} = k\frac{S_A - S_B}{D_{\text{inst}}^2 + \varepsilon^2}, \qquad \varepsilon > 0, \tag{35}$$

but regularisation is a modelling choice, not an established result.

10.3 Systemic importance

A body may be systemically important if omitting it changes predictions beyond a tolerance. For a dynamical system $X(t)$, define

$$E_j(T) = \sup_{0 \le t \le T} \|X_{\text{full}}(t) - X_{-j}(t)\| . \tag{36}$$

If $E_j(T) > \epsilon$, body j cannot be omitted under tolerance ϵ. This can be adapted to institutions: a central bank, payment network, energy system or observatory may be systemically important. Gravitational influence and financial contagion are not the same mechanism; the analogy concerns systemic dependence.

11 Biology, chemistry, ecology and planetary resilience

Biological organisms coordinate specialised subsystems through signalling, feedback, redundancy and repair. The analogy can inform resilience design but should not naturalise political hierarchy. Human institutions are governed by law, values and contestable authority; biological selection is not a political legitimacy principle.

A planetary response depends on material stocks and flows: energy carriers, water, metals, semiconductors, medicines, fertilisers and industrial feedstocks. Conservation laws constrain physical production. A financial allocation plan that ignores chemical inputs and industrial conversion capacity is incomplete.

Defensive or emergency action can damage ecosystems and create long-lived pollution. Environmental impact must be treated as a welfare and intergenerational constraint. The objective is not merely immediate survival but preservation of ecological systems on which future survival depends.

12 Psychology, psychiatry, linguistics and epistemic coordination

Threat perception can amplify availability bias, confirmation bias, group polarisation and motivated reasoning. Institutional communication should distinguish evidence from interpretation, use calibrated probability language and communicate uncertainty without exaggerating or concealing it.

Psychiatric terminology should not be used to stigmatise political disagreement or ordinary fear. Clinical claims require appropriate evidence and professional standards. Disagreement with an authority is not evidence of pathology.

A planetary information architecture must handle multilingual communication, semantic ambiguity, translation uncertainty and culturally different risk concepts. A common technical vocabulary can improve coordination, but translation must preserve uncertainty, qualifications and legally significant distinctions.

Let $T_{i,t} \in [0,1]$ denote receiver i's trust in a source. A stylised dynamic is

$$T_{i,t+1} = \text{clip}_{[0,1]}\left(T_{i,t} + \alpha_i A_{i,t} - \beta_i M_{i,t} - \gamma_i C_{i,t}\right), \tag{37}$$

where A denotes verified accuracy, M material misinformation and C unacknowledged correction failure. This is a proposed behavioural specification, not a universal empirical law.

13 Political science, geopolitics and diplomacy

States may resist cooperation if they fear unequal burdens, domination, free-riding or loss of future policy autonomy. Agreements must define contributions, decision rules, dispute resolution, exit provisions and protections for smaller participants.

In repeated interaction, cooperation may be sustained when future benefits, monitoring and credible remedies outweigh short-term gains from defection. Discount factors, enforcement capacity and distribution of gains matter. No single equilibrium is guaranteed across all parameter values.

Diplomacy provides mechanisms for signalling intentions, reducing misperceptions, negotiating common protocols and managing disagreement. Scientific communication and diplomatic communication should remain distinct but interoperable: scientific institutions assess evidence; authorised political institutions make accountable decisions.

14 Warfare economics, security and civil protection

The hypothetical external threat may require collective protection, but military action is not the only response. Monitoring, diplomacy, infrastructure hardening, evacuation planning, continuity of government, medical readiness and civil defence may dominate more escalatory options under many states of the world.

The analysis is strategic and institutional. It does not provide operational instructions for weapons construction, targeting or offensive violence.

Decision-makers should account for adversarial adaptation, accidental escalation, misinterpretation and irreversible harm. Robust policies should be evaluated against multiple plausible models rather than one fragile forecast. The precautionary principle can support protective monitoring and resilience, but does not automatically justify unlimited coercive measures.

The welfare function must include civilian casualties, displacement, health-system capacity, food security, energy access, privacy and distribution of emergency burdens. Humanitarian law and applicable domestic and international law constrain the means and methods of response.

15 Statistics, econometrics and empirical identification

15.1 Evidence and outcomes

PBP is a proposed framework. Its claims require measurable outcomes and comparisons. Relevant outcomes include speed and accuracy of belief updating, calibration and Brier scores, time to coordinated action, policy compatibility, distribution of costs and benefits, false-alarm and missed-threat rates, institutional trust and resilience under misinformation or outages.

15.2 Panel specification

For regional unit i and time t, a candidate panel model is

$$Y_{i,t} = \alpha_i + \tau_t + \beta_1 PBP_{i,t} + \beta_2 Exposure_{i,t} + \beta_3 Capacity_{i,t} + \boldsymbol{\gamma}' \mathbf{Z}_{i,t} + u_{i,t}. \tag{38}$$

Here Y might measure response latency, forecast calibration or policy compatibility. Fixed effects do not by themselves solve endogeneity. The design must address selection into information regimes, common shocks, spillovers, measurement error and interference between units.

15.3 Event studies and difference-in-differences

If a protocol is adopted at different times across regions, an event-study design may estimate dynamic associations. Causal interpretation requires plausible parallel trends, no anticipatory treatment effects, careful treatment of staggered adoption, and explicit modelling of spillovers. In a planetary network, standard no-interference assumptions are especially questionable.

15.4 Forecast evaluation

For binary events, the Brier score is

$$BS = \frac{1}{N} \sum_{n=1}^{N} (p_n - o_n)^2, \tag{39}$$

where p_n is the forecast probability and $o_n \in \{0, 1\}$ the outcome. Calibration, discrimination and decision usefulness should be reported separately. A well-calibrated forecast can still be unhelpful if it arrives too late or does not change feasible decisions.

The central causal question is not simply whether communication and convergence correlate. It is whether a specified information architecture changes beliefs, decisions and welfare relative to a credible counterfactual. Randomised experiments may be possible in low-stakes simulations; high-stakes planetary events require observational, quasi-experimental, laboratory and computational evidence with careful external-validity limits.

16 Computer science, algorithms, AI/ML and cybersecurity

A computational implementation may contain sensor ingestion and provenance validation; data fusion and uncertainty quantification; Bayesian state estimation; scenario simulation and robust optimisation; multi-agent coordination; human authorisation and legal review; immutable audit logs; and post-decision evaluation. AI systems should provide decision support, not unreviewable sovereign authority.

Let each agent i have local observation o_i, belief b_i, action a_i and policy π_i. The joint transition is

$$s_{t+1} \sim P(s_{t+1} \mid s_t, a_{1,t}, \ldots, a_{n,t}), \tag{40}$$

and each agent seeks to maximise a utility that may include local and shared components. Multi-agent reinforcement learning can explore protocols, but learned policies require interpretability, distribution-shift testing and human-controlled constraints.

16.1 Pseudocode

```
PROCEDURE Planetary_Bayesian_Coordination
    INPUT:
        observations from independent scientific sources
        prior beliefs and uncertainty models
        legal mandates and rights constraints
        regional capabilities and resource limits
        candidate information policies

    VERIFY source provenance, calibration, and integrity
    ESTIMATE state probabilities and model uncertainty
    PUBLISH evidence, assumptions, and dissenting assessments

    FOR each candidate information policy pi:
        COMPUTE receiver-specific posterior beliefs
        COMPUTE receiver best responses
        CHECK Bayes plausibility where common-prior assumptions apply
        CHECK legality, rights, and distributional constraints
        SIMULATE strategic interaction and adversarial scenarios
        ESTIMATE welfare, tail risk, and coordination compatibility
    END FOR

    PRESENT feasible policies to authorised decision-makers
    OBTAIN lawful human authorisation
    EXECUTE only approved and proportionate actions
    MONITOR outcomes, update beliefs, and preserve audit records
    TRIGGER independent review and sunset provisions
END PROCEDURE
```

Cybersecurity threats include sensor spoofing, data poisoning, model extraction, denial of service, compromised credentials, insider threats and supply-chain compromise. Controls include cryptographic provenance, independent replication, least privilege, segmented networks, secure update procedures, offline fallbacks, red-team exercises and incident disclosure. No cryptographic mechanism proves that a sensor's physical interpretation is true. Authenticity, integrity, scientific validity and strategic meaning are distinct properties.

17 Formal propositions and design principles

Assumption 17.1 (Common-prior benchmark)**.** *For the benchmark result only, receivers share a common prior over a finite state space and correctly understand the committed signal structure.*

Assumption 17.2 (Feasible action sets)**.** *Each receiver has a nonempty compact action set and continuous expected utility.*

Proposition 17.3 (Existence of a receiver best response)**.** *Under the feasible-action assumption, each receiver has at least one optimal action for every posterior belief.*

Proof. For a fixed posterior, expected utility is continuous in the action. A continuous function on a nonempty compact set attains its maximum by the extreme value theorem. □

Proposition 17.4 (Information does not guarantee agreement)**.** *Bayes-plausible information can change posterior beliefs without making all receivers choose the same action.*

Proof. Bayes plausibility constrains the ex ante average posterior, not equality of receiver utilities, action sets or realised posterior beliefs. If receivers have different payoff functions, their best responses may differ at the same posterior. If they have different priors or receive different signals, their posteriors may also differ. Agreement therefore does not follow from Bayesian updating alone. □

Proposition 17.5 (Institutional plurality can be compatible with coordination)**.** *If a nonempty set of joint actions satisfies shared safety, interoperability and legal constraints, multiple independent institutions do not by themselves preclude coordinated action.*

Proof. The constraints apply to the joint action profile. They need not impose a common legal identity, common objective function or identical policy instruments. A feasible profile can therefore be implemented by multiple independent institutions, provided the agreed interfaces and commitments are credible. □

Remark 17.6. *These propositions establish logical possibilities, not empirical claims that a particular planetary architecture will work. Existence of individual best responses does not prove existence or uniqueness of a strategic equilibrium in every multi-agent game. Additional conditions are required for such results.*

18 Financial and institutional optimisation

Let Π_t denote a partition of institutional functions among independent bodies, federations, shared services or consolidated entities. Let S_t collect the state of the system. A candidate recursive problem is

$$V(S_t) = \max_{\Pi_t \in \mathcal{F}(S_t)} \{W(S_t, \Pi_t) - K(\Pi_t) + \beta \mathbb{E}_t[V(S_{t+1})]\}. \tag{41}$$

Here W is a distribution-sensitive welfare measure, K includes transition, governance and operational costs, and $\mathcal{F}(S_t)$ is the legally and technically feasible set. Institutional design is endogenous; more consolidation is not assumed to be better.

Under deep uncertainty, reversible investments may have option value. A staged approach can preserve the ability to revise institutional arrangements as evidence improves. A robust architecture can be assessed by

$$\max_{\Pi} \min_{m \in \mathcal{M}} W_m(\Pi), \tag{42}$$

subject to legal and safety constraints. Robust optimisation does not eliminate uncertainty; it makes sensitivity to uncertainty explicit.

19 Implementation roadmap

Table 1: A proposed implementation roadmap.

Stage	Workstream	Deliverable and validation test
1	Definitions	Define functions, agents, states, actions, mandates and affected populations; compare alternative decompositions.
2	Evidence	Establish scientific provenance, data standards, uncertainty metadata and independent replication.
3	Legal design	Map jurisdiction, authorisation, rights, appeal, oversight and sunset provisions.
4	Monetary and fiscal	Model liquidity, solvency, payment continuity, real resource constraints and burden sharing.
5	Information design	Specify priors, signals, sender commitment, receiver responses and Bayes-plausibility conditions.
6	Simulation	Compare decentralised, federated and centralised protocols in multi-agent simulations and stress tests.
7	Welfare evaluation	Report means, quantiles, subgroup harms, rights constraints and intergenerational effects.
8	Pilot	Test low-stakes interoperability, crisis communications and audit protocols before high-stakes deployment.
9	Review	Publish failures, update models, preserve dissent and terminate ineffective or unlawful arrangements.

20 Falsifiable hypotheses

The framework generates hypotheses rather than settled findings:

1. Better provenance and independent verification improve forecast calibration and reduce the effect of spoofed signals.
2. Information policies that explicitly represent uncertainty improve decision quality relative to overconfident binary communication.
3. Interoperable regional protocols can improve response compatibility without requiring institutional merger.
4. Concentrating information and decision authority can improve speed in some scenarios but increase single-point-of-failure and legitimacy risks.
5. Distribution-sensitive objectives produce different policy choices from aggregate-welfare maximisation when burdens are unequal.
6. Independent scientific replication and audit reduce persistence of material information errors.
7. The six-function decomposition has measurable advantages only if it outperforms plausible five-, seven- or alternative-function architectures on pre-specified criteria.

Each hypothesis requires an operational definition, comparison group or model, pre-specified metric and explicit falsification condition.

21 Limitations

First, the six-node architecture is a proposed functional decomposition, not a mathematical necessity. Second, Bayesian persuasion assumes a model of beliefs and signal generation; misspecified priors, strategic senders and heterogeneous information can invalidate naive conclusions. Third, a welfare objective embeds normative choices that mathematics alone cannot resolve. Fourth, the Jupiter scenario is fictional and does not establish the existence of extraterrestrial threats. Fifth, the institutional-gravitational

analogy is not a physical law. Sixth, multi-agent simulations can reveal failure modes but cannot by themselves validate real-world causal effects. Seventh, AI/ML systems can amplify errors and power asymmetries unless constrained by law, audit and human accountability.

22 Conclusion

The Ghoshian Six-Body Problem is formulated here as a problem of coordinating six interdependent planetary functions under uncertainty. Its motivating numerical identity, $6 = 1 + 2 + 3$, is a structural mnemonic rather than a proof of optimal institutional cardinality. Its central innovation is to connect local monetary convergence with planetary coordination through a proposed multi-receiver information-design framework.

Planetary Bayesian Persuasion asks whether transparent, Bayes-consistent and accountable information structures can help heterogeneous institutions make compatible decisions without being forced into uniformity. The answer is not assumed. It must be established through theory, simulation, empirical evaluation, legal scrutiny and public contestability.

The Jupiter thought experiment makes the welfare question unavoidable: welfare of whom? The answer must include individuals, communities, nations, humanity, future generations and other affected sentient stakeholders where relevant. No institution should be treated as the ultimate beneficiary merely because it controls the coordination mechanism.

The resulting design principle is:

$$\boxed{\begin{array}{c}\text{Plural institutions,}\\ \text{shared evidence,}\\ \text{compatible decisions,}\\ \text{lawful authority,}\\ \text{distribution-sensitive welfare.}\end{array}} \tag{43}$$

Local convergence is one problem. Planetary convergence is another. Solar System convergence is a further horizon. The task is not to presume that each scale must be governed by one institution, but to discover which functions must be coordinated, which should remain plural, and how the welfare of all affected populations can be protected as interdependence expands.

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Glossary

Bayes plausibility
The condition that posterior beliefs average to the prior before the signal is observed.

Bayesian persuasion
Information design in which a sender commits to a signal structure to influence a receiver's beliefs and actions.

Body
A functional component of a coupled institutional architecture; not necessarily a physical celestial body.

Compatibility
The degree to which distinct institutional actions can operate together under shared constraints.

Epistemic security
Protection of the authenticity, integrity, provenance and interpretability of information.

Institutional convergence
Increasing alignment or interoperability of institutional practices, rules or functions.

Institutional gravitational pull
A proposed analogy for convergence pressure arising from differences in institutional capability or stability.

Legitimacy
The justified acceptability of authority and decisions among those subject to or affected by them.

Planetary Bayesian Persuasion
The proposed multi-receiver extension of information design for accountable planetary coordination.

Planetary convergence
Coordination among regional and global systems without necessarily requiring institutional uniformity.

Receiver
An agent that observes a signal, updates beliefs and chooses an action.

Sender
An agent that selects or commits to an information structure.

Six-body architecture
The proposed six-function system of observation, verification, deliberation, finance, logistics and welfare oversight.

Welfare
A multidimensional assessment of effects on affected persons and populations, including distribution and future effects.

The End